



ADVENTURE WORK SALES REPORT



ABSTRACT

Adventure Works (fictional company) supplies sports products and accessories across the globe. This sales report is generated on the basis of Power BI dashboard for Adventure Works sales data, integrating executive-level KPIs (revenue, profit, orders, returns), customer segmentation via logistic regression, and trend forecasting. Provide insightful visuals including decomposition trees, seasonality matrix, and key influencers using AI-driven analytics to uncover sales drivers. Enabled data-driven strategic decisions by analyzing regional sales performance, product trends, and customer behavior, resulting in actionable insights to improve revenue and marketing effectiveness.

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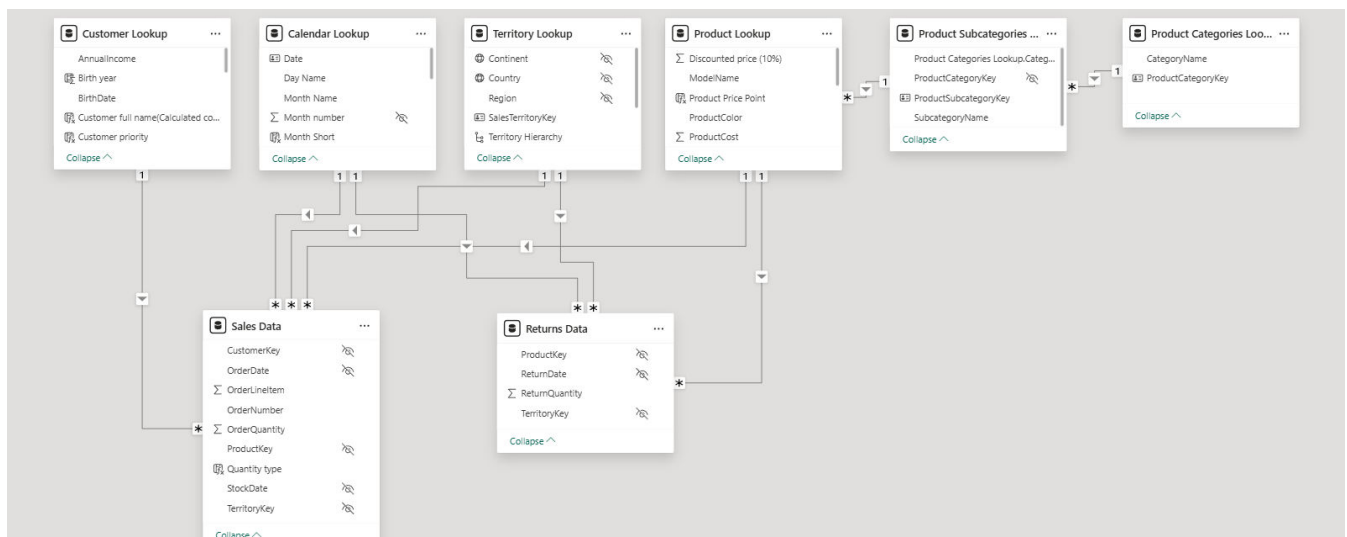


Adventure Works Business Report

1. Objective

This report enables Adventure Works' sales and marketing teams to track sales performance, returns, product insights, and customer trends across various regions, time periods, and product categories. Below underlying data model integrates key business dimensions to provide a unified and interactive view of operational metrics.

2. Data Model Overview



The data model follows a star schema structure, optimized for analytical queries and visualizations. The central fact tables are:

2.1 Sales Data – Contains transactional sales metrics such as:

- OrderDate, OrderQuantity, StockDate, OrderNumber
- ProductKey, CustomerKey, TerritoryKey

2.2 Returns Data – Tracks returned products with:

ProductKey, ReturnQuantity, ReturnDate, TerritoryKey

These fact tables are connected to various dimension tables in one-to-many relationship.



The **dimension tables** are listed below:

Dimension Table	Key Attributes
Customer Lookup	Annual Income, Birth Year, Full Name, Customer Priority
Calendar Lookup	Date, Month Name, Day Name, Month Number, Month Short
Product Lookup	Product Model, Color, Cost, Discounted Price
Product Subcategories	Subcategory and associated Product Category Key
Product Categories Lookup	Category Names (e.g., Bikes, Clothing, Accessories)
Territory Lookup	Region, Country, Continent

Key Relationships:

- One-to-many from each dimension to the fact tables
 - Relationships are based on primary keys (e.g., ProductKey, CustomerKey)
 - Separate lookup (stage table) layers for product category hierarchy for snowflake schema to support many-to-many relationship (Product Subcategories and Product Categories Lookup)
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3. Dashboard Highlights Overview

3.1. Sales Overview

- Total Sales by Region, Product Category, and Time
- Sales Trends using time-based visuals (Monthly, Quarterly, Yearly)
- Top Products by Revenue or Volume Sold

3.2. Returns Analysis

- Return Rate as a percentage of total orders
- Analysis of product categories and regions with highest return volumes
- Trend of returns over time to identify product or logistics issues

3.3. Customer Demographics

- Sales segmented by customer income levels, birth year, and priority status
- Identification of high-value customers

3.4. Territory Performance

- Geographic sales patterns across countries and continents
- Comparative performance by Sales Territory

Business Insights Derived

- Certain product categories experience higher return rates, prompting quality checks.
 - Sales growth is strongest in specific months, indicating seasonal demand.
 - High-income customers show a preference for premium product lines.
 - Underperforming territories can be targeted for marketing campaigns or resource allocation.
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4. [Dashboard Highlights Detail](#)

4.1. Executive Summary Dashboard

Objective

This executive-level dashboard provides a snapshot of overall business performance for Adventure Works. It highlights key metrics and visual trends to support strategic decision-making by leadership, sales, and operations teams.

2. Dashboard Features and KPIs

Key Performance Indicators (KPIs)

Metric	Value	Description
Total Revenue	\$25M	Overall revenue generated from product sales
Total Profit	\$10.46M	Net profit after deducting costs
Total Orders	25K	Total number of customer orders fulfilled
Return Rate	2.17%	Proportion of products returned relative to total products sold

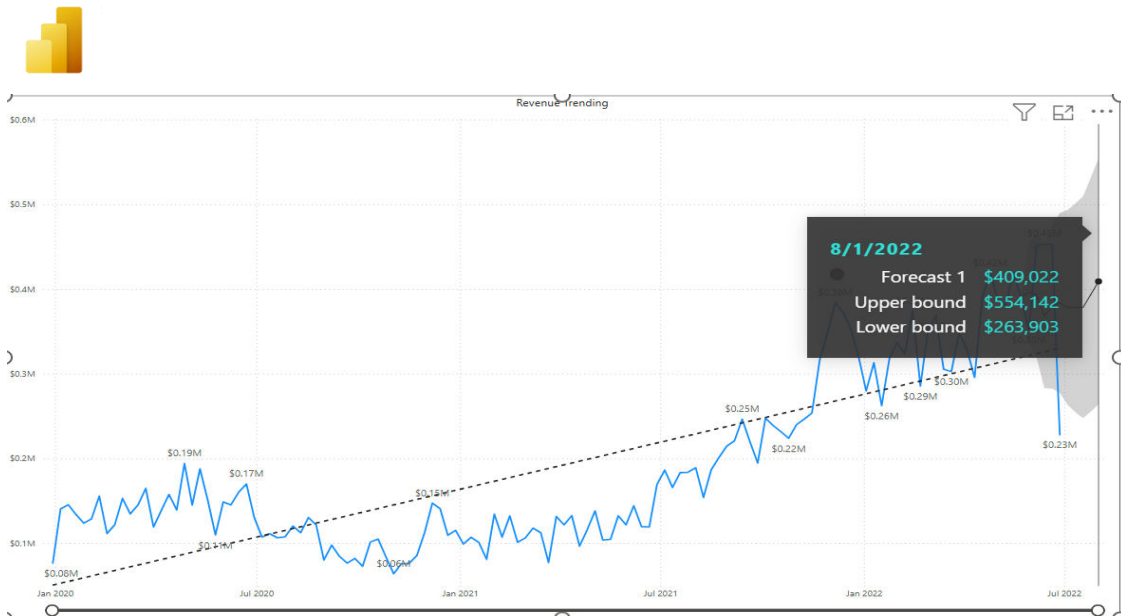
Monthly KPIs

- Monthly Revenue: \$1.83M (+3.31% from previous month)
- Monthly Orders: 2,146 (-0.88% from previous month)
- Monthly Returns: 166 (+1.78% from previous month)

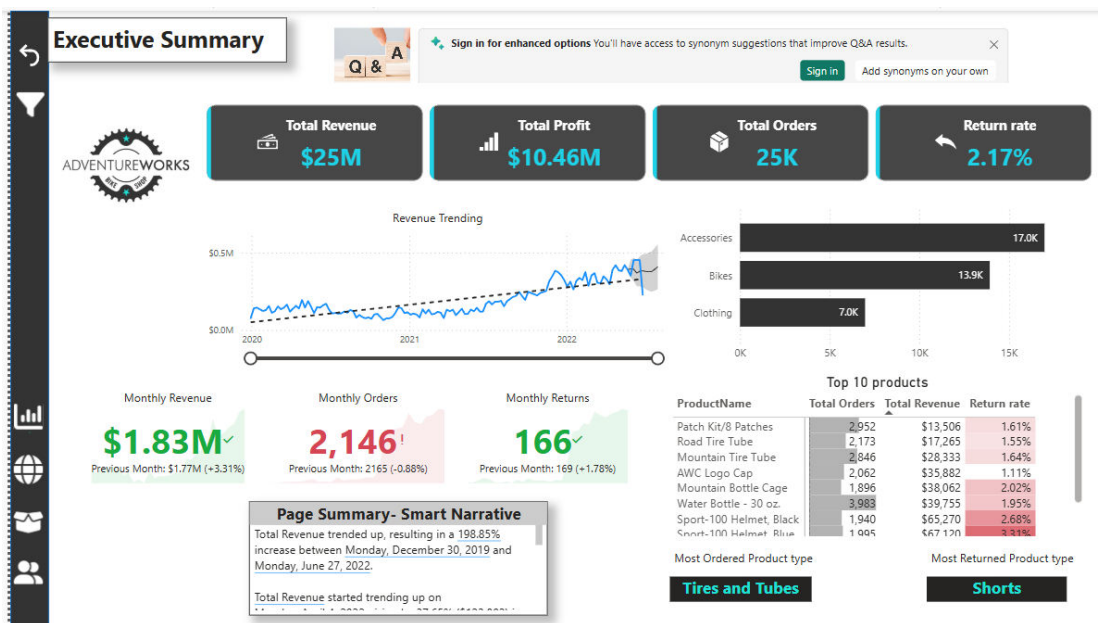
3. Data Visualizations and Analytical Features

Revenue Trend with Forecast

- A time series line chart displaying revenue performance from 2020 to 2022.
- Forecasting applied: A forecast section for August 1, 2022, provides estimated revenue of Forecasted value \$409,022 (Upper Bound: \$554,142 and Lower Bound: \$263,903) with 95% Confidence interval.



- Trendline with confidence intervals helps visualize revenue trajectory.



Product Category Breakdown

- Horizontal bar chart comparing total products sold across:
 - Accessories: 17.0K units
 - Bikes: 13.9K units
 - Clothing: 7.0K units



Top 10 Products Table

- Ranked by Total Orders, with additional metrics:
 - Revenue generated per product
 - Return rate per product (highlighted with conditional formatting)
- Key Insight: Tires and Tubes are most ordered; Shorts have the highest return rate.

Smart Narrative Summary

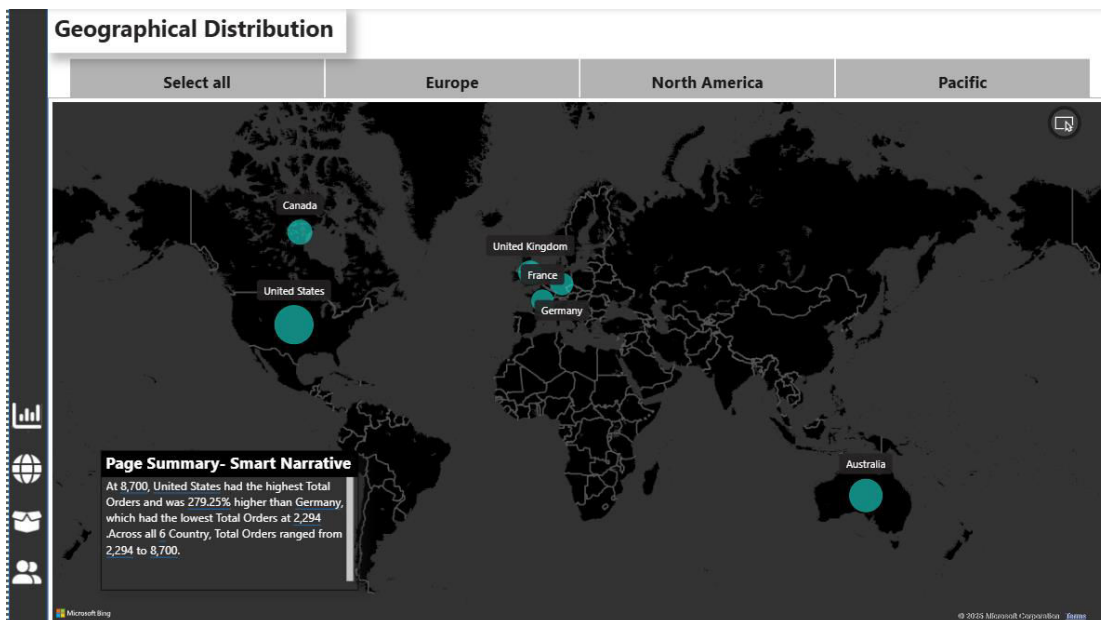
- AI-generated summary (Smart Narrative) that dynamically interprets data for business users:

"Total Revenue trended up, resulting in a 198.85% increase between Dec 30, 2019 and Jun 27, 2022."

5. Business Insights & Recommendations

- **Growth Trend:** A consistent increase in revenue indicates strong sales performance and growing market presence.
- **Return Monitoring:** Return rate is low overall, but specific SKUs like Sport-100 Helmet, Blue (3.31%) may warrant quality review.
- **Product Focus:** Accessories dominate volume; promoting high-performing items like Tires and Tubes may drive profitability.

4.2. Geographical Distribution – Total Orders by Region





Objective

This dashboard page visualizes total customer orders across six countries, offering regional insights to support marketing, logistics, and territory sales strategies.

Global Map View

A location-based bubble map (Bing-powered) illustrates order volumes by country:

Country	Total Orders	Insight
United States	8,700	Highest order volume; leads global sales with 279.25% more orders than Germany.
Germany	2,294	Lowest number of orders across all regions analyzed.
Australia	Mid-range	Strong Pacific presence; potential for deeper market penetration.
United Kingdom	-	Stable volume; clusters well with other European nations.
France	-	Moderate presence in continental Europe.
Canada	-	Supports North American operations; second to the U.S. in this region.

Smart Narrative Insights

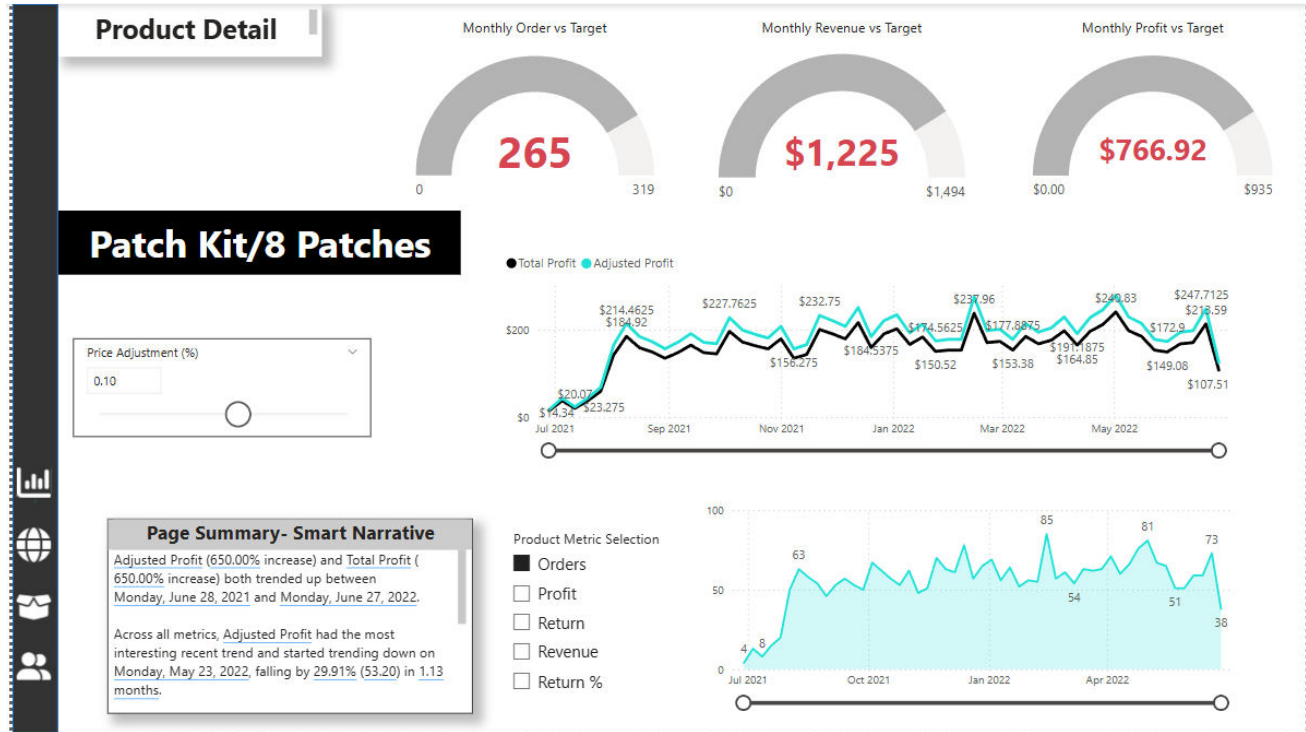
"At 8,700, United States had the highest Total Orders and was 279.25% higher than Germany, which had the lowest Total Orders at 2,294. Across all 6 countries, Total Orders ranged from 2,294 to 8,700."

Strategic Takeaways

- **Focus Market:** The U.S. remains the dominant market—marketing and logistics investments should be optimized here.
- **Growth Opportunities:** Consider campaigns or regional partnerships in Germany and France to boost their comparatively low order volumes.
- **Regional Diversification:** Australia's strong performance indicates potential for Pacific expansion strategies.



4.3. Product Performance Dashboard



Objective

To offer an in-depth view of the top-selling product (*Patch Kit/8 Patches*) using performance KPIs, historical trends, and profit forecasting via price adjustment simulation.

KPI Performance vs Target

Metric	Value	Target	Gap Analysis
Monthly Orders	265	319	Below target (-54)
Monthly Revenue	\$1,225	\$1,494	-18%
Monthly Profit	\$766.92	\$935	-18%

These metrics highlight a shortfall in performance for the current period, though not significantly off track.



Price Sensitivity Analysis

- **Price Adjustment Slider (%)**: Enables scenario-based planning by adjusting discounts or price increases. This can help in taking strategic decisions to decide how much discount can be given to boost the profit in future.
 - **Insight**: A 0.10% adjustment (10% discount) shows its potential impact on *Adjusted Profit*, aiding decision-making around promotions or price revisions.
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Profit Trend (Jul 2021 – Jun 2022)

- Total Profit vs Adjusted Profit is plotted to show how profit shifts with minor price changes.
 - **Peak Performance**:
 - March 2022 saw the highest adjusted profit (\$237.96).
 - **Recent Dip**:
 - From May 23, 2022, *Adjusted Profit* dropped by 29.91% in just 1.13 months.
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Performance Trend by Metrics

The bottom area chart allows toggling between:

- Orders
- Profit
- Return
- Revenue
- Return %

E.g., Orders rose significantly from 4 to 63 between July and October 2021, peaking at 85 in early 2022.

Smart Narrative Summary

"Adjusted Profit (650.00% increase) and Total Profit (650.00% increase) both trended up between June 28, 2021, and June 27, 2022. However, Adjusted Profit fell by 29.91% starting May 23, 2022."



Strategic Takeaways

- **Target Refinement:** Product is performing close to target; slight marketing or pricing tweaks may help achieve KPIs.
- **Profit Optimization:** Price adjustments are highly influential—this product is sensitive to discounting.
- **Inventory & Promotion:** Increased order spikes and returns must be considered in stock and campaign planning.

4.4. Customer Insight & Segmentation Analysis

Objective

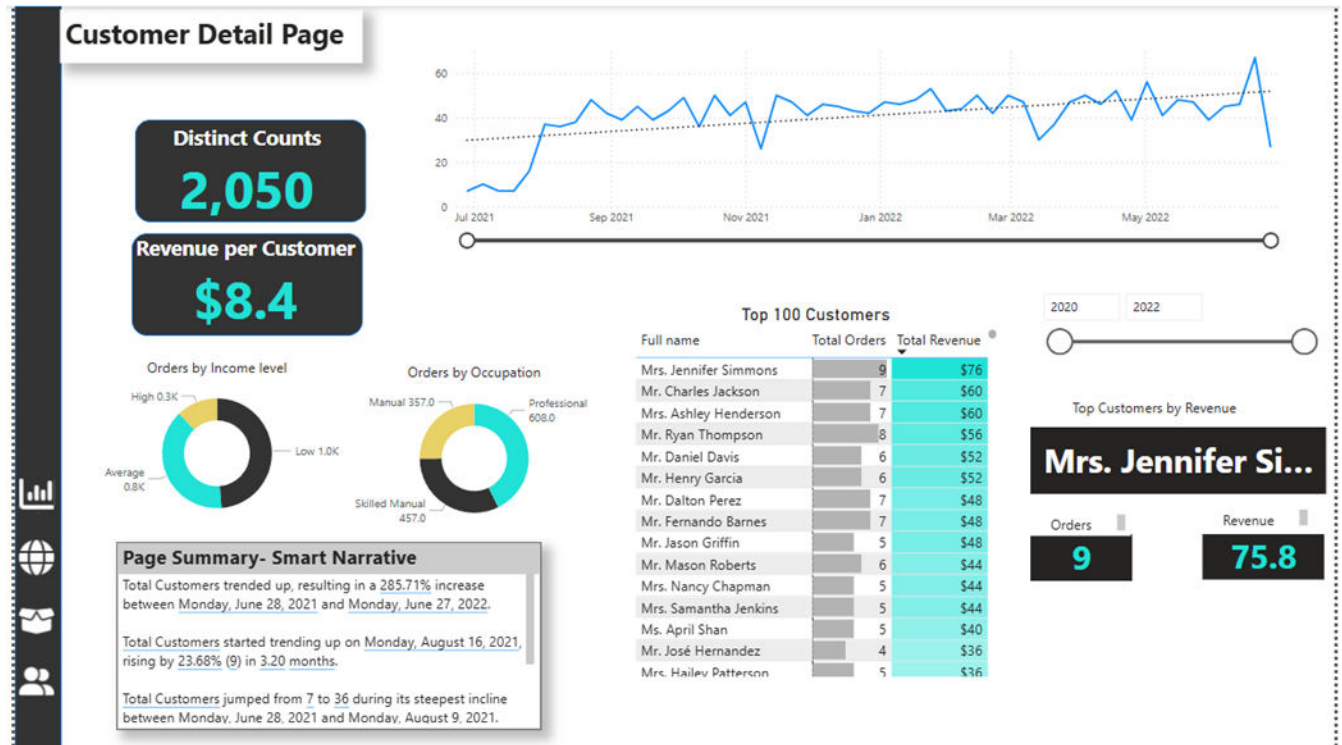
This dashboard provides a comprehensive view of the customer base, enabling strategic decision-making through customer segmentation, behavior analysis, and revenue tracking from key clients.

Key Highlights

Metric	Value
Distinct Customers	2,050
Revenue per Customer	\$8.4
Top Customer	Mrs. Jennifer Simmons (9 Orders, \$76 Revenue)

Customer Growth Trends

- **Total Growth:** 285.71% increase in customer base from June 28, 2021 to June 27, 2022.
 - **Notable Surge:** From 7 to 36 customers within 6 weeks (Jun 28 to Aug 9, 2021).
 - **Recent Uptick:** 23.68% growth starting August 16, 2021, over 3.2 months.
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Customer Segmentation

By Income Level

- Average Income Customers: 0.8K orders (largest group)
- Low Income: 1.0K orders
- High Income: 0.3K orders

By Occupation

- Professional: 608 orders (largest segment)
- Skilled Manual: 457 orders
- Manual: 357 orders

Insight: Professionals and average-income customers are the most active buyers. Marketing campaigns can be tailored accordingly for better ROI.

Top 100 Customers

- Sorted by Total Orders and Revenue
- Mrs. Jennifer Simmons leads with 9 orders, contributing \$76



- Others like Mr. Ryan Thompson and Mrs. Ashley Henderson follow with 7–8 orders

Actionable Strategy

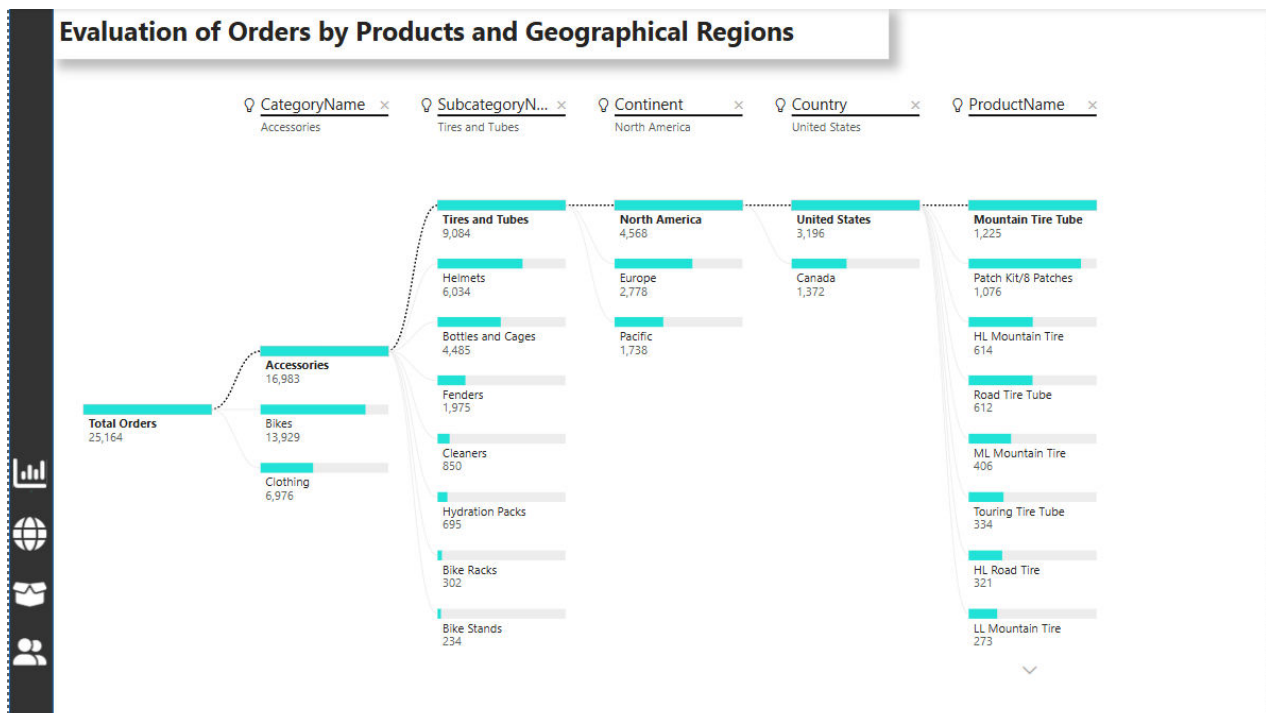
- High-value Customers: Reward with loyalty discounts or early access
- Low-engagement Customers: Use targeted deals to increase activity

Strategic Takeaways

- Segment-Based Marketing: Classify customers based on income and job role to personalize engagement.
- Churn Prevention: Monitor declining activity among mid-tier customers.
- Customer Lifetime Value (CLV): Focus on increasing revenue per customer from \$8.4 through upselling and bundling.

4.5. Root Cause Analysis & Strategic Recommendations for Business Growth

We leveraged data-driven dashboards to uncover patterns, evaluate product performance, and understand customer behavior to support actionable business decisions. This analysis is structured into three main dimensions:



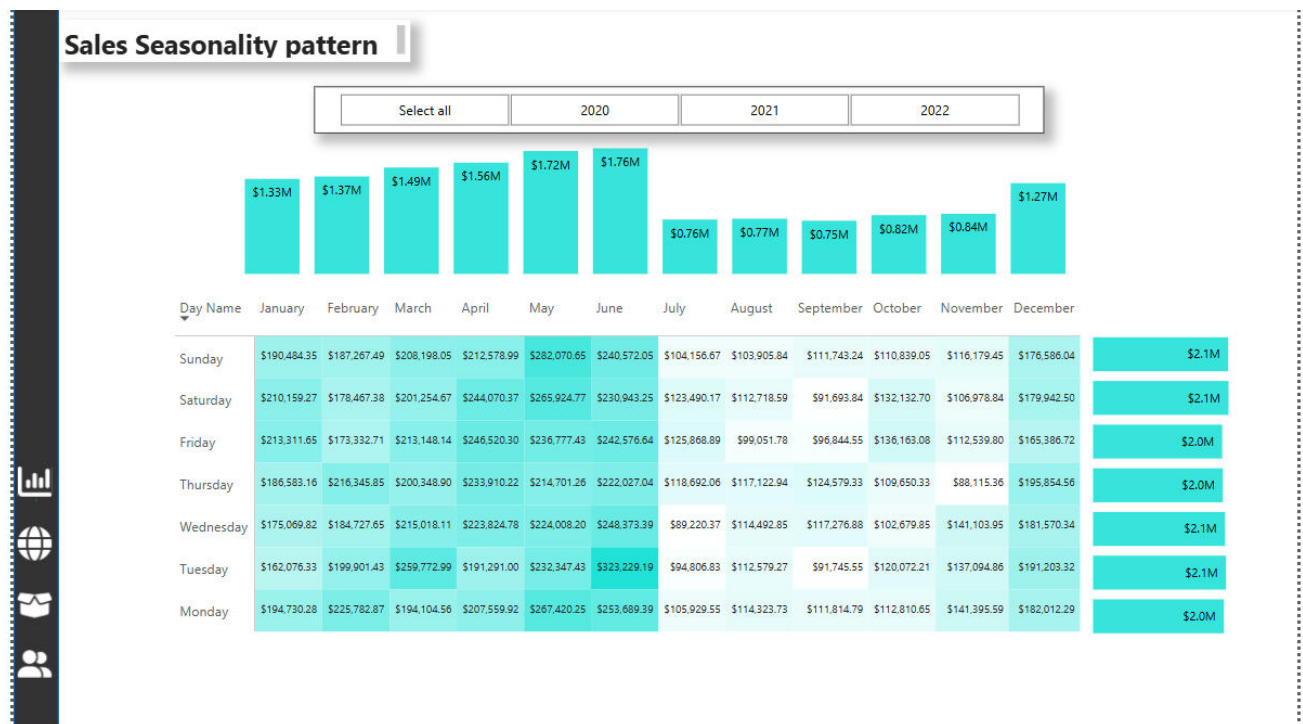


1. Evaluation of Orders by Products and Geographical Regions

Using a hierarchical drill-down from product category to country-level granularity:

- High and Low Selling Products: The "Accessories" category, particularly *Tires and Tubes*, is a top performer. However, regional differences emerge—e.g., the *United States* leads in orders, with *Mountain Tire Tube* being the top product.
- **Strategic Implications:**
 - Focus marketing and supply chain resources on high-demand products in strong markets.
 - Investigate low-selling regions for potential gaps in awareness, logistics, or customer fit.
 - Consider region-specific product bundles or promotions.

2. Sales Seasonality Pattern



The temporal analysis uncovers strong monthly and weekday trends:

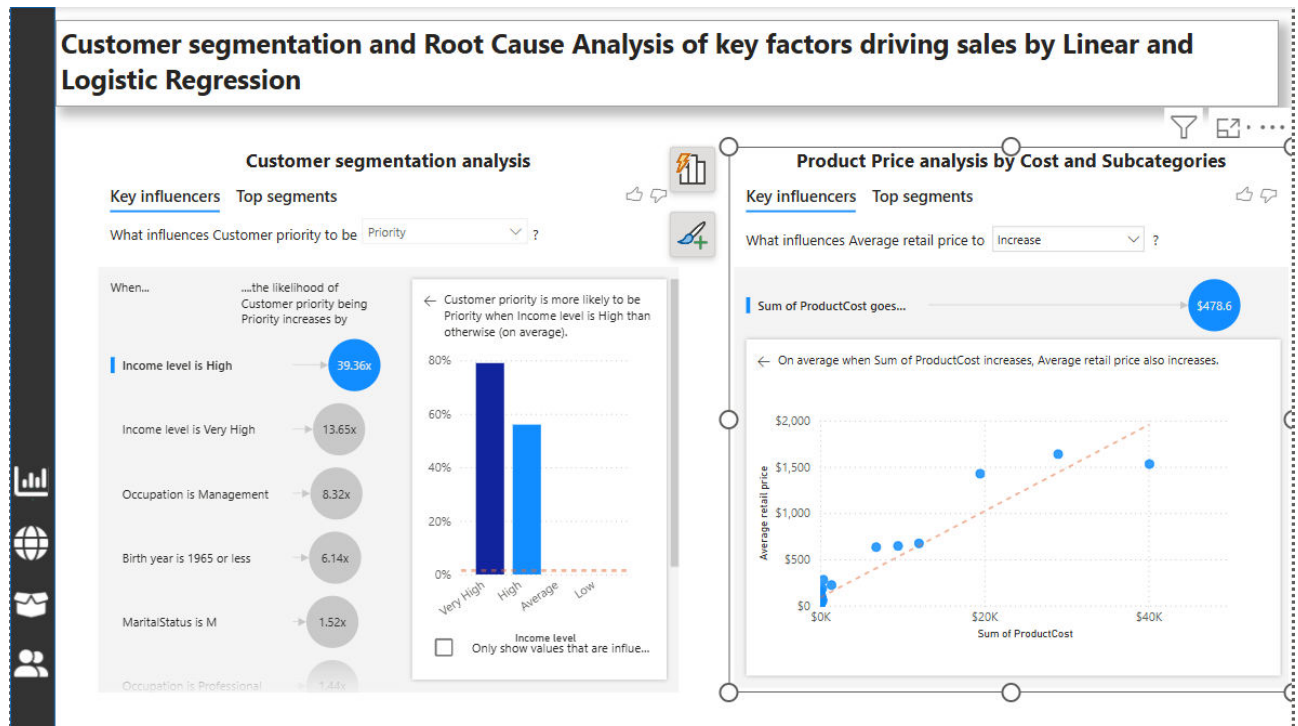
- Peak Sales: May and June are top months, with *Tuesday to Saturday* showing consistently higher daily revenues.
- Dip in Sales: July to November shows a clear decline, suggesting seasonal fatigue or off-peak demand.



- **Strategic Implications:**

- Launch major campaigns and product drops in Q2 (April–June).
- Introduce discounts or loyalty programs during off-peak months to sustain engagement.
- Adjust inventory and staffing based on expected demand curves.

3. Customer Segmentation & Sales Drivers (Regression-Based Analysis)



Through Linear and Logistic Regression models, we derived key drivers of sales performance:

- **Customer Segmentation:**

- *Income Level* is the most significant predictor—customers with higher income are 39.36x more likely to be tagged as priority.
- Professional and management roles further correlate with higher purchasing tendencies.

- **Product Pricing:**

- There is a clear positive correlation between product cost and average retail price—products with higher cost bases tend to command higher prices.



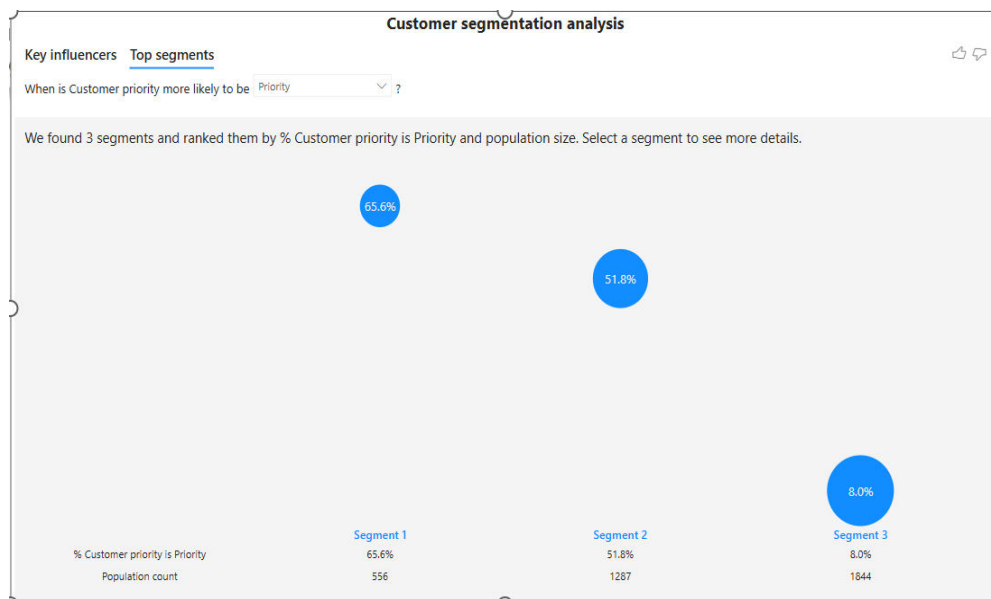
- **Strategic Implications:**

- Prioritize affluent customer segments through personalized promotions and premium offerings.
- Implement value-based pricing strategies—leveraging cost insights for optimized margins.

Customer segmentation analysis using logistic regression. Customers can be grouped into priority and standard category.

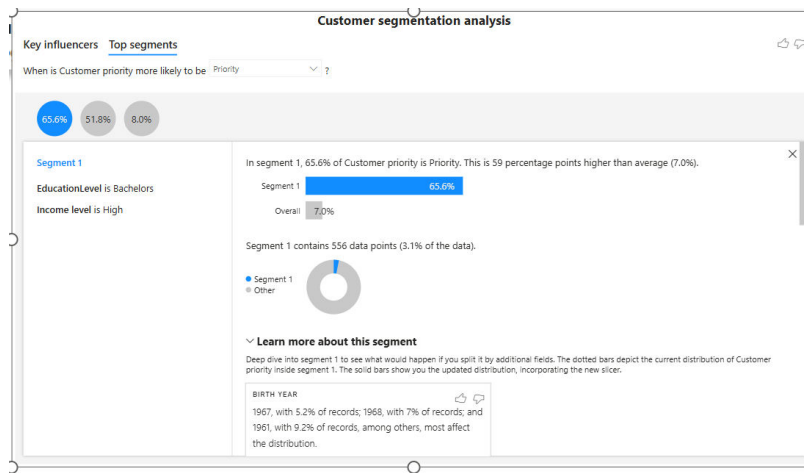


In priority, customers are grouped into 3 segments.

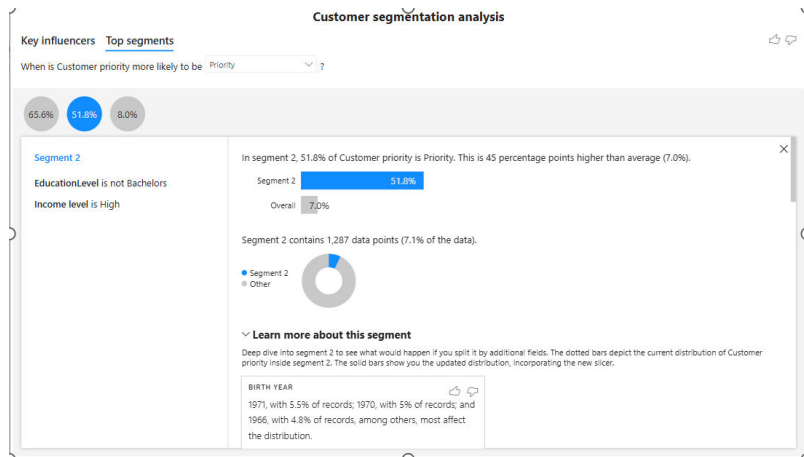




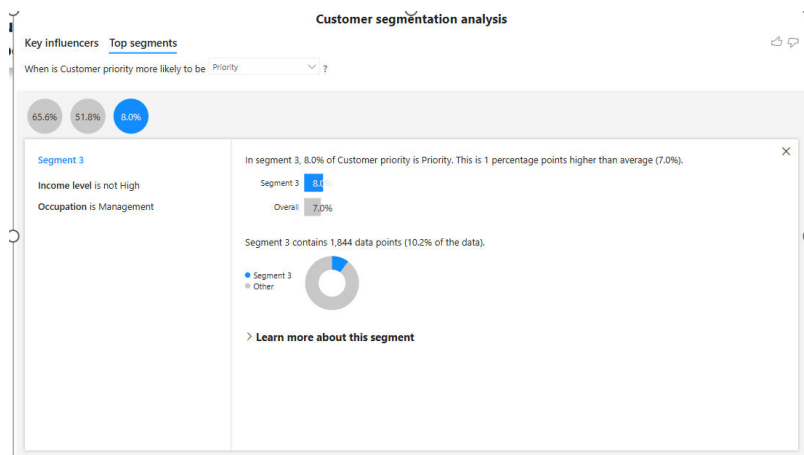
Customer Segment-1: Education level is Bachelors; income level is high.



Customer Segment-2: Education level is not-bachelors and income level are high.



Customer Segment-3: Income is not high and occupation is management.





5. Conclusion

The Power BI report for Adventure Works provides a 360-degree view of sales performance, customer behavior, and product trends. Through segmentation, forecasting, and influencer modeling, the report empowers data-driven strategic decisions in:

- Marketing personalization
- Product lifecycle management
- Regional expansion
- Revenue forecasting
- Customer retention

This end-to-end analysis, powered by Power BI, becomes a decision support system for business growth, operational efficiency, and customer satisfaction. This Report Drives Business Strategy by following ways-

1. **Customer-Centric Decision Making:**
 - a. Target high-value customers using segmentation insights.
 - b. Improve loyalty with personalized outreach.
2. **Sales Optimization:**
 - a. Forecasting enables inventory planning and financial projections.
 - b. Key influencer analysis directs focus toward revenue-generating factors.
3. **Market Expansion:**
 - a. Geo analytics help identify potential growth regions.
 - b. Data supports decisions on market entry or exit.
4. **Product Strategy:**
 - a. Understand product profitability to drive innovation and inventory control.
 - b. Phase out underperforming SKUs.
5. **Operational Efficiency:**
 - a. Identify bottlenecks and root causes with decomposition trees.
 - b. Use predictive modeling to prevent churn and optimize pricing.